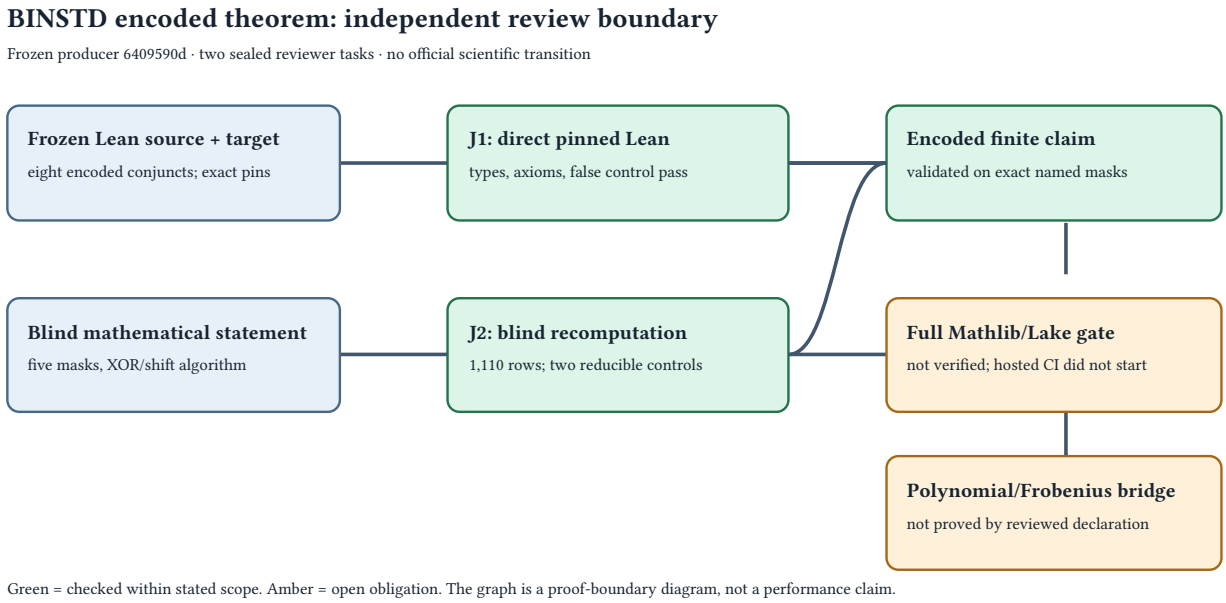
BINSTD encoded certificate: independent review

26 October 2026 · producer `6409590d` · plan `cab536aa`

Two sealed reviewers examined separate joints. J1 checked the exact frozen Lean declaration, direct pinned compilation, seven types, seven axiom outputs and a rejected false control. J2 independently enumerated 1,110 monic candidate rows from the statement alone. The repository independence checker passed after separate compatibility addenda were filed; original attestations and the initial format failure are retained.



Claim results

Scope	Machine	Outcome
Five exact encoded masks; 510/30/30 candidate counts	passed	validated scoped
Frozen Lean declaration; direct compile/type/axioms	passed direct	inconclusive as full certificate
Archived direct-check receipt	passed replay	validated direct scope
Full Mathlib/Lake target gate	not verified	inconclusive
Polynomial irreducibility and Frobenius bridge	not verified	inconclusive

The direct Lean checks used pinned 4.35.0-rc1. Six axiom queries reported only `propext`; the count theorem had none. The false-control attempt failed as expected. Both reviewers agree on all five finite predicate values and first zero divisors. Reducible masks 131073 and 257 both divide by mask 3 with zero remainder, requiring 17 and 8 XOR steps.

Boundary and next proof

The theorem proves finite equalities of a source-defined bitmask function. It does not prove that function equals `(ZMod 2)[X]` remainder, factor irreducibility, or a normal-basis/Frobenius correspondence. The pinned full workspace gate was not run; hosted Actions executed no steps due an account payment or spending-limit gate. Run the exact target gate with exclusive host ownership, then prove the mask-to-polynomial and remainder bridges. No official scientific status changes here.

Evidence: `REPORT.md`, `archive/claim-verdicts.json`, `reconciliation.json`, `independence-check.json` and the two `reports/TASK-*/` directories in this packet. Review tasks: `TASK-20261026-538444`, `TASK-20261026-7ab9cb`.